Companion LXX: Wavelet Transforms of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

Wavelet transforms of persistence diagrams provide a joint scale–position representation of topological fluctuations in cosmological fields. In weak-lensing analyses, wavelet coefficients quantify how the birth and death of connected components, loops, and void-like structures vary across both scale and filtration parameter. In the standard cosmological model, the wavelet structure reflects the nonlinear matter distribution and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the multi-scale topology in a characteristic way. This Companion develops a continuous description of wavelet transforms in RDCC, deriving how the relaxation scale influences multi-scale topological coherence, wavelet-energy distributions, and the observational signatures in weak-lensing surveys. The resulting framework provides a distinctive probe of the RDCC gravitational field through multi-scale topology.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. While spectral density functions provide a frequency-resolved view of topological fluctuations, wavelet transforms offer a joint representation in scale and position, capturing localized multi-scale structure.

In the standard cosmological model, the wavelet structure reflects the nonlinear matter distribution and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the multi-scale topology in a scale-dependent manner, producing a distinctive signature in wavelet analyses.

2 Wavelet Transforms of Persistence Diagrams

Given a filtration-ordered sequence of persistence diagrams $\{D(\nu)\}$, the continuous wavelet transform (CWT) of the Wasserstein-distance fluctuations is

$$W_D(a,b) = \int W_p(D(\nu), D^*)\psi^*\left(\frac{\nu - b}{a}\right)\frac{d\nu}{\sqrt{|a|}}, \tag{1}$$

where a is the scale, b is the position, and ψ is the mother wavelet.

The wavelet energy is

$$E_D(a) = \int |W_D(a,b)|^2\, db. \tag{2}$$

3 RDCC Modification of Persistence Diagrams

In RDCC, the convergence evolves as

$$\kappa_{\mathrm{RDCC}}(\ell) = \kappa(\ell)\left[1 - \chi_\kappa\left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^\alpha\right]. \tag{3}$$

This modifies the birth and death thresholds of topological features:

$$b_{\mathrm{RDCC}} = b\left[1 - \chi_b(\ell/\ell_{\mathrm{rel}})^\alpha\right], \tag{4}$$

$$d_{\mathrm{RDCC}} = d\left[1 - \chi_d(\ell/\ell_{\mathrm{rel}})^\alpha\right]. \tag{5}$$

Thus the RDCC wavelet coefficients become

$$W_{D,\mathrm{RDCC}}(a,b) = W_D(a,b)\left[1 - \chi_W\left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^\alpha\right]. \tag{6}$$

4 Wavelet Energy in RDCC

The RDCC wavelet energy is

$$E_{D,\mathrm{RDCC}}(a) = E_D(a)\left[1 - \chi_E\left(\frac{\ell}{\ell_{\mathrm{rel}}}\right)^\alpha\right]. \tag{7}$$

This modification reflects the relaxation-driven change in multi-scale topology.

5 Multi-Scale Topological Signatures of RDCC

The relaxation mechanism enhances the coherence of large-scale modes. Wavelet transforms therefore exhibit:

- enhanced wavelet energy at large scales,
- suppressed wavelet energy at small scales,
- a characteristic turnover near ℓ_{rel},
- modified scale–position localization of topological features,
- altered alignment of persistence lifetimes across scales.

These signatures provide a direct probe of the RDCC gravitational field through multi-scale topology.

6 Conclusion

Wavelet transforms of persistence diagrams in the Relaxation-Driven Cyclic Cosmology reflect the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the multi- scale topology in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of multi-scale topology, nonlinear structure, and the interplay between light and gravity in RDCC.

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